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SUPERSEDING
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MILITARY SPECIFICATION

CARTRIDGES, OR SHOT, FIXED: HVAP-T: M93A1 FOR 76-MM GUNS; M304 FOR 90-MM GUNS; T66E3 AND T66E4 FOR 76-MM GUN T91; M332A1, M332A1B1 FOR 90-MM GUNS, M3 AND T119, HVTP-T: M315A2 FOR 76-MM GUN M1; M317A2 FOR 90-MM GUNS; M333A1 FOR 90-MM GUNS, M3 AND T119

1. SCOPE

1.1 This specification covers assembling, marking, and packing of armor-piercing, and target-practice hypervelocity cartridges, and fixed shot (hereinafter referred to as a round) for use in 76-mm and 90-mm guns.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, drawings, and publications, and all specifications and drawings, referenced thereon, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

MIL-G-2550	-General Specification For Ammunition Except Small Arms Ammunition

DRAWINGS

ORDNANCE CORPS

75-1-220	-Cartridge, HVAP-T, M93A1, Smokeless, For 76-MM Guns. Assembly And Marking Diagram.
75-1-234	-Shot, Fixed, HVAP-T, M304, Smokeless, For 90-MM Guns. Assembly And Marking Diagram.
75-1-249	-Shot, Fixed, HVTP-T, M315A2, For 76-MM Gun, M1. Assembly And Marking Diagram.
75-1-262	-Shot, Fixed, HVTP-T, M317A2, Smokeless, For 90-MM Guns. Assembly And Marking Diagram.
75-1-295	-Cartridge, HVAP-T, T66E3 And T66E4, MV4135 For 76-MM Gun, T91. Assembly And Marking Diagram.
75-1-310	-Shot, Fixed, HVAP-T, M332A1,M332A1B1 And Shot, Fixed, HVTP-T, M333A1, For 90-MM Gun M3 And T119. Assembly And Marking Diagram.

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PUBLICATIONS

ORDNANCE CORPS

ORD-SIP-M11-102 -Cartridge, HVAP-T And HVTP-T For 76-MM And 90-MM Guns. Loaded

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable specifications and drawings.

3.2 The round shall comply with all requirements specified on Drawings 75-1-220, 75-1-234, 75-1-249, 75-1-262, 75-1-295, or 75-1-310, as applicable, and all drawings referenced thereon, and with all requirements specified in applicable specifications.

3.3 Propellant

3.3.1 76-mm rounds.-The weight of the propellant shall not differ by more than 0.15 ounce avoirdupois from that designated by the loading authorization for the particular lot of propellant used.

3.3.2 90-mm rounds.-The weight of the propellant shall not differ by more than 0.2 ounce avoirdupois from that designated by the loading authorization for the particular lot of propellant used.

3.3.3 Weighing of charge.-The contractor shall weigh and checkweigh each propellant. The checkweighing shall be accomplished by an operator working independently of the original weigher, using a different scale from that used to make the first weighing.

3.4 Igniter charge.-The weight of an individual igniter charge, where applicable, shall not vary from that specified on the applicable drawing by more than 0.1 ounce.

3.5 Crimp

3.5.1 76-mm rounds.-The cartridge case shall separate from the shot at a load between 2,000 pounds minimum and 8,500 pounds maximum when tested as specified in 4.4.

3.5.2 90-mm rounds.-The cartridge case shall separate from the shot at a load between 5,000 pounds minimum and 11,000 pounds maximum when tested as specified in 4.4.

3.6 Proving ground requirements for 76-mm and 90-mm rounds

3.6.1 Average velocity.-The difference between the velocities of any two rounds fired in the velocity test shall not exceed 3 percent of the mean velocity of all the rounds fired. The average velocity shall not differ from the specified velocity (see 6.2) by more than 2 percent after reducing results to standard conditions, when tested as specified in 4.5.

3.6.2 Tracer functioning.-The tracer shall not burn less than 3 seconds when tested as specified in 4.5.

3.6.3 Security.-There shall be no indication of separation of the metal parts in the gun bore or in flight when tested as specified in 4.5.

3.7 Restoration of painting.-The painting and marking shall be restored by the assembling contractor on any components furnished to him painted and marked, when such painting and markings have been marred or obliterated.

3.8 Workmanship

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3.8.1 All parts shall be free from chips, dirt, grease, rust, and other foreign material. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent used.

3.8.2 Care shall be exercised at all times so that the rotating band of each projectile is not damaged by being nicked or burred.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot

4.1.1 A lot shall consist of all rounds assembled by one manufacturer under one contract, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Each lot shall contain:

a. brass or steel cartridge cases of new manufacture made by one manufacturer in accordance with the same drawing, and same drawing revision, same specification, and same specification revision.

b. propellant of same lot.

c. loaded primers from not more than two lots.

d. igniting charges of same lot.

e. armor-piercing shot from not more than one lot.

f. target practice shot from not more than one lot.

g. rounds of one type only.

4.1.2 Small lots.-Small lots containing fewer than 1,000 rounds shall be grouped in lots of 1,000 to 2,000 for the purpose of selection of ballistic samples, and shall be produced on one production line, by the same manufacturing method, except that such grouping shall be restricted to lots having the same lot of propellant.

4.2 Sampling.-Number of samples, acceptance, and rejection criteria used for determination of lot acceptance for the test specified in 4.4 shall be in accordance with Standard Inspection Procedure ORD-SIP-M11-102. Fifteen rounds shall be selected by the inspector for proving ground tests specified in 4.5.

4.3 Inspection.-Inspection shall be as specified in Specification MIL-G-2550.

4.3.1 Gaging.-After the round has been assembled, the contractor shall gage each round of ammunition in a profile and alignment gage (commonly known as a chamber gage) pertaining to this round. If the round fails to enter the profile and alignment gage freely, it shall be rejected subject to correction and retest. Drawings of Government profile and alignment gage will be furnished to the contractor for his information upon application to the inspector.

4.4 Test of crimping case to shot

4.4.1 Procedure.-Sample rounds shall be tested in a satisfactory tensile testing machine, to determine the security of the crimp of the cartridge case to the shot, by applying a load in the axial direction. Rate of travel of testing machine head shall be 0.100 to 0.125 inch per minute inclusive. The load at which the cartridge case separates from the shot shall be observed to determine compliance with 3.5.1, or 3.5.2, , as applicable.

4.5 Proving ground test of the round

4.5.1 Procedure.-Five sample rounds shall be fired from a gun for which the round is standard. The time of tracer functioning shall be determined by a stop watch or other suitable means and the velocity shall be measured to determine compliance with 3.6.1 and 3.6.2. Observation shall be made for separation of metal parts to determine compliance with 3.6.3.

4.5.2 Acceptance.-The lot shall be considered acceptable provided the requirements of 3.6.1, 3.6.2 and 3.6.3 are met. The lot shall be considered eligible for retest provided no metal parts separate from the projectiles and if:

a. the difference between the velocities of any two rounds fired in the velocity test exceeds 3 but does not exceed 4 percent of the mean velocity

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of all rounds fired or,

b. the average velocity differs from the prescribed velocity by more than 2 percent, but not more then 3 percent, or

c. two tracers burn less than 3 seconds.

If the limits for eligibility for retest are exceeded, the lot shall be rejected.

4.5.3 Retest.-A retest as provided for in 4.5.2 may be made if requested by the contractor using double the number of samples used in the original test. The lot shall be considered acceptable on retest provided the requirements of 3.6.1, 3.6.2 and 3.6.3 are met.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, labeling, and marking.- Packaging, packing, labeling and marking shall be as specified on the applicable drawing.

5.2 Data cards.-Data card information shall be as specified in Specification MIL-G-2550.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number, and date of this specification.

6.2 Information concerning the specified velocity may be obtained from the procuring agency.

6.3 When the contractor for the round is also the contractor for one or more of the components thereof, the proving ground tests of the components may be combined with the proving ground tests of the round to save expense, upon agreement between the United States and the contractor to that effect. In the event the round specification does not cover all of the proving ground tests specified for the components, the additional tests prescribed in the component specification should be conducted.

Notice.-This specification, together with specifications and drawings pertaining to it and bearing a "Notice" of similar restrictions, is intended for use only in connection with procurement by the United States Government; and shall not be reproduced either wholly or in part except when authorized in connection with Government procurement, or be used for any other purpose except when specifically authorized by the Chief of Ordnance.

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